BENG 285/BNFO 285/ECE 204. Statistical Learning in Bioinformatics

Project 2: Detecting enrichments and depletions of rare events within large datasets.

Learning objectives: The overall learning objective of this project is to utilize statistical learning approaches for recognizing events which are more or less likely to happen purely by chance. The project will require for students to construct background models reflecting the null hypothesis and to test whether the provided data are consistent or inconsistent with this hypothesis. Students will likely need to consider a multitude of null hypotheses necessitating the need for additional statistical corrections. It is further anticipated that students will need to utilize both nonparametric and/or parametric statistical tests for the completion of the project. Lastly, students will be required to compare their analysis to existing state-of-the-art approaches for detecting rare events.

Subject Matter: The focus on project 2 will be the detection of positively, negatively, and neutrally mutated genes in cancer genomes. While the genome of a cancer can have many thousands (and even millions) of somatic mutations, most cancers will have less than 10 mutations which are under positive and/or negative selection. Positively selected mutations are advantageous for a cancer, and they increase its survival and proliferation. From a statistical perspective, positively mutated genes should be mutated more in cancer genomes than purely expected by chance. Negatively selected mutations are disadvantageous for a cancer and result in its death or senescence. From a statistical perspective, negatively mutated genes should be mutated less than purely expected by chance. Neutrally selected mutations are ones that have no effect on the tumor's fitness. From a statistical perspective, there should be no difference between the expected and observed mutation rates of these genes. Each team will be provided with a dataset of somatic mutations found in cancer genomes and they need to identify the set of positively, negatively, and neutrally selected genes. Students will also need to compare their results to the results from applying existing tools to their respective datasets.

Project's manuscript: https://www.nature.com/articles/s41568-020-0290-x
Manuscript presentation: Team presentation on Tuesday (22-Apr) at 6:30pm (CENTR 217B)
Project's datasets: https://www.dropbox.com/scl/fo/7d37xqur5vlb8jni61b0t/AClf0-cbGkGBSZFIs9fXzdQ?rlkey=pfw7xmb7slnz7d398gzfzpju7
Project's presentation: Team presentation on Thursday (1-May) at 6:30pm (CENTR 217B)
Deadline: All teams submit a project report (max 3 pages) by 11:59pm on Friday (2-May)

Potential outline for approaching the project:

- Identify the number of somatic mutations within each gene.
- Perform corrections for the size of the gene and the expected mutation rate in that gene.
- Evaluate the functional significance of the observed mutations within each gene.
- Using statistical comparisons, determine whether the observed mutations are likely affecting the function of each gene more or less than purely expected by chance.
- Be careful of multiple hypothesis testing issues and consider reducing the number of initial hypotheses.
- Compare your results to the ones from directly applying existing tools (such as dNdScv, OncodriveFML, CBaSE, OncodriveCLUSTL, and/or others) to the provided dataset.